

### 3.1.5

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## 1 3.1.5 — Cartan / SN derivator özdeşlikleri (form + dual multivector)

**Hedef.** 3.1.4'ün üç türetilmiş özdeşliği gibi, ama bu sefer *derivator-of-derivation* yapısında: bir derivasyon  $D$  ve bir bracket  $[\cdot, \cdot]$  ikilisi için *derivator* obstruction  $D_D^{[\cdot, \cdot]}(a, b) := D[a, b] - [Da, b] - (-1)^{|a||D|}[a, Db]$  sıfırdan farklı kalan altı kanonik kurguyu (üçü Koszul-bracket'i form tarafında, üçü Schouten–Nijenhuis bracket'i multivector tarafında) sentaktik olarak kapatmak:

#### Form

- (1)  $D_{\tilde{\mathcal{L}}_U}^{T^*M}(\eta, \mu) = \mathcal{L}_{\tilde{K}_U} \mu + K_{\tilde{K}_U} \eta,$
- (2)  $\mathcal{L}_{\tilde{d}_{\tilde{\iota}_U}} \mu = -[d_{\iota_U} \eta, \mu]_K,$
- (3)  $0 = d_{\iota_{\tilde{\mathcal{L}}_W}} \eta - d_{\tilde{d}_{\tilde{\iota}_W}} \omega + d_{\iota_W} [\omega, \eta]_K.$

#### Dual (multivector)

- (1')  $\tilde{D}_{\tilde{\mathcal{L}}_U}^{\text{SN}}(U, V) = \tilde{\mathcal{L}}_{K_U} V + \tilde{K}_{K_U} U,$
- (2')  $\tilde{\mathcal{L}}_{d_{\iota_U}} V = -[\tilde{d}_{\tilde{\iota}_U} U, V]_{\text{SN}},$
- (3')  $0 = \tilde{d}_{\tilde{\mathcal{L}}_U} V - \tilde{d}_{\tilde{d}_{\tilde{\iota}_U}} U + \tilde{d}_{\tilde{\iota}_U} [U, V]_{\text{SN}}.$

Tu/Cattaneo §3.1.5'in altı özdeşliği bu çiftleme — (1)  $\leftrightarrow$  (1'), (2)  $\leftrightarrow$  (2'), (3)  $\leftrightarrow$  (3') — Koszul ile Schouten–Nijenhuis bracket arasındaki form/multivector dualitesini somutlaştırıyor. Tüm kanıtlar Poisson varsayımı altında ( $[\pi, \pi]_{\text{SN}} = 0$ ).

### 1.1 Strateji

**Form tarafı (1)/(2)/(3).** `KoszulProblem.derivator_form_engine()` — `intrinsic_engine_with_closure` tabanı + form-side Cartan intrinsics + 10 closure aksiyomu (Faz 15.C pass 1) — `prove_derivator(..., side="form")` üzerinden  $Y$  bir vector field probe'u ile değerlendirilen `MultiEval` sarması altında kapanıyor.

**Multivector tarafı (1')/(2')/(3').** `derivator_multivector_engine()` form tarafının dual genişlemesi: aynı tabana 13 closure aksiyomu (Faz 15.C pass 2 + pass 3) ekleniyor:

- $\text{SN} \rightarrow \text{LBVF}$  coercion (1-vfs için), tilde-iota / pairing-flip / d-on-scalar / L-on-scalar bridge'leri, HVF iç-iota normalizasyonu, SN/LBVF Neg ve Sum doğrusallıkları (pass 2),
- `TildeLieOnOneVectorLichnerowiczDefinition` —  $\tilde{\mathcal{L}}_\alpha U = -X_{\alpha(U)} + \pi^\sharp(\mathcal{L}_U \alpha) - [U, \pi^\sharp \alpha]_{VF}$  Lichnerowicz açılımı,

- `SharpNegLinearityDefinition` —  $\pi^\sharp(\text{Neg}(A)) \rightarrow \text{Neg}(\pi^\sharp(A))$  böylece işaretler dışarı çıkıp eşleşip iptal olabiliyor,
- `BareAtomSlotLiftDefinition` — `LieBracketVF` gibi opak operator atomlarının özel slotları, atom doğrudan `MultiEval` argümanı olduğunda da inner-engine ile açılıyor (pass 3).

`prove_derivator(..., side="multivector")`  $\xi$  bir 1-form probe'u ile (`slot_kind="covector"`) değerlendiriyor.  $Y$  ve  $U/V/W$ 'nin ayarları `KoszulProblem` ile çözülüyor:  $Y, Z$  probe'lar `Derivation` atom;  $U, V, W$  ise registry-declare edilmiş `Symbol` 1-vector — bu sayede `_is_plain_vf` ve closure aksiyomları doğru şekilde tanyor.

```
[1]: # Notebook doğrudan açıldığında gradalg'ı import edilebilir hâle getirir.
try:
    import gradalg # noqa: F401
except ModuleNotFoundError:
    import sys
    from pathlib import Path
    here = Path.cwd().resolve()
    for candidate in (here, *here.parents):
        if (candidate / "gradalg" / "__init__.py").is_file():
            sys.path.insert(0, str(candidate))
            break
    import gradalg # noqa: F401
```

## 1.2 1. Kurulum

Tek bir `KoszulProblem` üzerinden hem form hem multivector tarafını test ediyoruz: form inventory ( $\omega, \eta, \mu$ ) 1-formlar + multivector inventory ( $U, V, W$ ) 1-vfs. Probe değişkenleri:

- $Y, Z$  — `Derivation` atomları (form-side `MultiEval` evaluasyonu),
- $\xi, \zeta$  — registry-declare edilmiş 1-form `Symbol`'leri (multivector-side `MultiEval` evaluasyonu).

`prob.assume_poisson()` Jacobi koşulunu varsayıyor; tilde tarafının  $\tilde{d}^2 = 0$ ,  $\tilde{\mathcal{L}}_\pi = 0$  vs. kuralları bu varsayım üzerinden tetikleniyor.

```
[2]: from gradalg.algebra.derivation import Act, Derivation
from gradalg.brackets.base import BracketApply
from gradalg.brackets.schouten import sn as default_sn
from gradalg.calculus.cartan_remainder import K
from gradalg.calculus.derivator import derivator
from gradalg.calculus.exterior_d import d as default_d
from gradalg.calculus.interior import interior
from gradalg.calculus.lie_derivative import lie_derivative
from gradalg.calculus.tilde import K_tilde, tilde_d, tilde_interior, tilde_lie
from gradalg.core.expr import Neg, Sum, Symbol
from gradalg.core.properties import Graded
from gradalg.core.registry import PropertyRegistry
from gradalg.display.jupyter import display_chain
from gradalg.library.koszul_problem import KoszulProblem
```

```

reg = PropertyRegistry()
pi = Symbol("π")
omega = Symbol("ω"); reg.declare(omega, Graded(degree=1))
eta = Symbol("η"); reg.declare(eta, Graded(degree=1))
mu = Symbol("μ"); reg.declare(mu, Graded(degree=1))
U = Symbol("U"); reg.declare(U, Graded(degree=1))
V = Symbol("V"); reg.declare(V, Graded(degree=1))
W = Symbol("W"); reg.declare(W, Graded(degree=1))

Y = Derivation("Y", 0)
Z = Derivation("Z", 0)
xi = Symbol("ξ"); reg.declare(xi, Graded(degree=1))
zeta = Symbol("ζ"); reg.declare(zeta, Graded(degree=1))

prob = KoszulProblem(
    pi, (omega, eta, mu),
    registry=reg,
    multivectors=((U, 1), (V, 1), (W, 1)),
)
prob.assume_poisson()
K_b = prob.koszul_bracket

print("Form-side engine kural sayısı      :",
      len(prob.derivator_form_engine().definitions))
print("Multivector-side engine kural sayısı:",
      len(prob.derivator_multivector_engine().definitions))

```

```

Form-side engine kural sayısı      : 33
Multivector-side engine kural sayısı: 63

```

## 1.3 2. Form tarafı

**1.3.1** (1)  $D_{\mathcal{L}_U}^{T^*M}(\eta, \mu) = \mathcal{L}_{\tilde{K}_\eta U} \mu + K_{\tilde{K}_\mu U} \eta$

Sol taraf  $\mathcal{L}_U$  derivasyonunun Koszul bracket  $[\eta, \mu]_K$  üzerindeki obstruction'ı; sağ tarafta  $\tilde{K}$  (tilde Cartan-remainder) ile bağlanan iki düzeltme terimi.  $Y$  ile değerlendirilip skalere düşülüyor.

```

[3]: lhs = derivator(lie_derivative(U), K_b, eta, mu)
K_tilde_eta_U = Act(K_tilde(eta, pi), U)
K_tilde_mu_U = Act(K_tilde(mu, pi), U)
rhs = Sum(
    Act(lie_derivative(K_tilde_eta_U), mu),
    Act(K(K_tilde_mu_U), eta),
)
chain = prob.prove_derivator(lhs, rhs, eval_args=(Y,), side="form")
print(f"(1) form-side, D^K_{{L_U}}(eta,mu) = L_{{Ktilde_eta U}} mu +_
      ↪K_{{Ktilde_mu U}} eta --> {len(chain)} adimda kapandi")

```

`display_chain(chain)`

(1) form-side,  $D^*K_{\{L_U\}}(\eta, \mu) = L_{\{K\tilde{\eta} U\}} \mu + K_{\{K\tilde{\mu} U\}} \eta$   
 $\rightarrow 109$  adimda kapandi

[3] :

$$\begin{aligned}
& [\eta, \mu]_{[\cdot, \cdot]_K[\pi]} \rightarrow L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(\langle \pi\sharp(\eta), \mu \rangle) \quad [[\alpha, \beta]_K = L_{\rho}\alpha(\beta) - L_{\rho}\beta(\alpha) - d\langle \rho\alpha, \beta \rangle]_{[\cdot, \cdot]_K[\pi]} \text{ (axiom)} \\
& \quad \langle \pi\sharp(\eta), \mu \rangle \rightarrow \mu(\pi\sharp(\eta)) \quad [\langle \alpha, X \rangle \rightarrow \alpha(X)] \text{ (axiom)} \\
& \quad \mu(\pi\sharp(\eta)) \rightarrow -\eta(\pi\sharp(\mu)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha))] [\pi] \text{ (axiom)} \\
& [L_U(\eta), \mu]_{[\cdot, \cdot]_K[\pi]} \rightarrow L_{\pi\sharp(L_U(\eta))}(\mu) - L_{\pi\sharp}(\mu)(L_U(\eta)) - d(\langle \pi\sharp(L_U(\eta)), \mu \rangle) \quad [[\alpha, \beta]_K = L_{\rho}\alpha(\beta) - L_{\rho}\beta(\alpha) - d\langle \rho\alpha, \beta \rangle]_{[\cdot, \cdot]_K[\pi]} \text{ (axiom)} \\
& \quad \langle \pi\sharp(L_U(\eta)), \mu \rangle \rightarrow \mu(\pi\sharp(L_U(\eta))) \quad [\langle \alpha, X \rangle \rightarrow \alpha(X)] \text{ (axiom)} \\
& \quad \mu(\pi\sharp(L_U(\eta))) \rightarrow -(L_U(\eta))(\pi\sharp(\mu)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha))] [\pi] \text{ (axiom)} \\
& (L_U(\eta))(\pi\sharp(\mu)) \rightarrow U(\eta(\pi\sharp(\mu))) - \eta([U, \pi\sharp(\mu)]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& [\eta, L_U(\mu)]_{[\cdot, \cdot]_K[\pi]} \rightarrow L_{\pi\sharp}(\eta)(L_U(\mu)) - L_{\pi\sharp(L_U(\mu))}(\eta) - d(\langle \pi\sharp(\eta), L_U(\mu) \rangle) \quad [[\alpha, \beta]_K = L_{\rho}\alpha(\beta) - L_{\rho}\beta(\alpha) - d\langle \rho\alpha, \beta \rangle]_{[\cdot, \cdot]_K[\pi]} \text{ (axiom)} \\
& \quad \langle \pi\sharp(\eta), L_U(\mu) \rangle \rightarrow (L_U(\mu))(\pi\sharp(\eta)) \quad [\langle \alpha, X \rangle \rightarrow \alpha(X)] \text{ (axiom)} \\
& (L_U(\mu))(\pi\sharp(\eta)) \rightarrow U(\mu(\pi\sharp(\eta))) - \mu([U, \pi\sharp(\eta)]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad \mu(\pi\sharp(\eta)) \rightarrow -\eta(\pi\sharp(\mu)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha))] [\pi] \text{ (axiom)} \\
& (L_U(L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(-\eta(\pi\sharp(\mu)))) - (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{\pi\sharp}(\mu)(L_U(\eta)) - d(-(U(\eta(\pi\sharp(\mu)))) - \eta([U, \pi\sharp(\mu)]_{VF}))) - (L_{\pi\sharp}(\eta)(L_U(\mu)) - L_{\pi\sharp(L_U(\mu))}(\eta) - d(-\eta(\pi\sharp(\mu))))(Y) \rightarrow U((L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(-\eta(\pi\sharp(\mu))))(Y)) - (L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(-\eta(\pi\sharp(\mu))))([U, Y]_{VF}) \\
& (L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(-\eta(\pi\sharp(\mu))))(Y) \rightarrow (L_{\pi\sharp}(\eta)(\mu))(Y) - (L_{\pi\sharp}(\mu)(\eta))(Y) - (d(-\eta(\pi\sharp(\mu))))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\eta)(\mu))(Y) \rightarrow (\pi\sharp(\eta))(\mu(Y)) - \mu([\pi\sharp(\eta), Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\eta), Y]_{VF} \rightarrow -[Y, \pi\sharp(\eta)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \mu(-[Y, \pi\sharp(\eta)]_{VF}) \rightarrow -\mu([Y, \pi\sharp(\eta)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\mu)(\eta))(Y) \rightarrow (\pi\sharp(\mu))(\eta(Y)) - \eta([\pi\sharp(\mu), Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\mu), Y]_{VF} \rightarrow -[Y, \pi\sharp(\mu)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \eta(-[Y, \pi\sharp(\mu)]_{VF}) \rightarrow -\eta([Y, \pi\sharp(\mu)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(-\eta(\pi\sharp(\mu))))(Y) \rightarrow Y(-\eta(\pi\sharp(\mu))) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& (L_{\pi\sharp}(\eta)(\mu) - L_{\pi\sharp}(\mu)(\eta) - d(-\eta(\pi\sharp(\mu))))([U, Y]_{VF}) \rightarrow (L_{\pi\sharp}(\eta)(\mu))([U, Y]_{VF}) - (L_{\pi\sharp}(\mu)(\eta))([U, Y]_{VF}) - (d(-\eta(\pi\sharp(\mu))))([U, Y]_{VF}) \quad [\text{MultiEval h}] \\
& (L_{\pi\sharp}(\eta)(\mu))([U, Y]_{VF}) \rightarrow (\pi\sharp(\eta))(\mu([U, Y]_{VF})) - \mu([\pi\sharp(\eta), [U, Y]_{VF}]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\eta), [U, Y]_{VF}]_{VF} \rightarrow -[[U, Y]_{VF}, \pi\sharp(\eta)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \mu(-[[U, Y]_{VF}, \pi\sharp(\eta)]_{VF}) \rightarrow -\mu([[U, Y]_{VF}, \pi\sharp(\eta)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\mu)(\eta))([U, Y]_{VF}) \rightarrow (\pi\sharp(\mu))(\eta([U, Y]_{VF})) - \eta([\pi\sharp(\mu), [U, Y]_{VF}]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\mu), [U, Y]_{VF}]_{VF} \rightarrow -[[U, Y]_{VF}, \pi\sharp(\mu)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \eta(-[[U, Y]_{VF}, \pi\sharp(\mu)]_{VF}) \rightarrow -\eta([[U, Y]_{VF}, \pi\sharp(\mu)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(-\eta(\pi\sharp(\mu))))([U, Y]_{VF}) \rightarrow [U, Y]_{VF}(-\eta(\pi\sharp(\mu))) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{\pi\sharp}(\mu)(L_U(\eta)) - d(-(U(\eta(\pi\sharp(\mu)))) - \eta([U, \pi\sharp(\mu)]_{VF}))) (Y) \rightarrow (L_{\pi\sharp(L_U(\eta))}(\mu))(Y) - (L_{\pi\sharp}(\mu)(L_U(\eta)))(Y) - (d(-(U(\eta(\pi\sharp(\mu)))) - \eta([U, \pi\sharp(\mu)]_{VF}))) (Y) \\
& (L_{\pi\sharp(L_U(\eta))}(\mu))(Y) \rightarrow (\pi\sharp(L_U(\eta)))(\mu(Y)) - \mu([\pi\sharp(L_U(\eta)), Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(L_U(\eta)), Y]_{VF} \rightarrow -[Y, \pi\sharp(L_U(\eta))]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \mu(-[Y, \pi\sharp(L_U(\eta))]_{VF}) \rightarrow -\mu([Y, \pi\sharp(L_U(\eta))]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\mu)(L_U(\eta)))(Y) \rightarrow (\pi\sharp(\mu))((L_U(\eta))(Y)) - (L_U(\eta))(\pi\sharp(\mu), Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& (L_U(\eta))(Y) \rightarrow U(\eta(Y)) - \eta([U, Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\mu), Y]_{VF} \rightarrow -[Y, \pi\sharp(\mu)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& (L_U(\eta))(-[Y, \pi\sharp(\mu)]_{VF}) \rightarrow U(\eta(-[Y, \pi\sharp(\mu)]_{VF})) - \eta([U, (-[Y, \pi\sharp(\mu)]_{VF})]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad \eta(-[Y, \pi\sharp(\mu)]_{VF}) \rightarrow -\eta([Y, \pi\sharp(\mu)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& \quad [U, (-[Y, \pi\sharp(\mu)]_{VF})]_{VF} \rightarrow -[U, [Y, \pi\sharp(\mu)]_{VF}]_{VF} \quad [(-X), Y]_{VF} \rightarrow -[X, Y]_{VF} \text{ (axiom)} \\
& \quad \eta(-[U, [Y, \pi\sharp(\mu)]_{VF}]_{VF}) \rightarrow -\eta([U, [Y, \pi\sharp(\mu)]_{VF}]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(-(U(\eta(\pi\sharp(\mu)))) - \eta([U, \pi\sharp(\mu)]_{VF}))) (Y) \rightarrow Y(-(U(\eta(\pi\sharp(\mu)))) - \eta([U, \pi\sharp(\mu)]_{VF})) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& (L_{\pi\sharp}(\eta)(L_U(\mu)) - L_{\pi\sharp(L_U(\mu))}(\eta) - d(U(-\eta(\pi\sharp(\mu)))) - \mu([U, \pi\sharp(\eta)]_{VF}))(Y) \rightarrow (L_{\pi\sharp}(\eta)(L_U(\mu)))(Y) - (L_{\pi\sharp(L_U(\mu))}(\eta))(Y) - (d(U(-\eta(\pi\sharp(\mu)))) - \mu([U, \pi\sharp(\eta)]_{VF}))(Y) \\
& (L_{\pi\sharp}(\eta)(L_U(\mu)))(Y) \rightarrow (\pi\sharp(\eta))((L_U(\mu))(Y)) - (L_U(\mu))(\pi\sharp(\eta), Y]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)}
\end{aligned}$$

$$\begin{aligned}
& (L_U(\mu))(Y) \rightarrow U(\mu(Y)) - \mu([U, Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \quad [\pi\sharp(\eta), Y]_{VF} \rightarrow -[Y, \pi\sharp(\eta)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& (L_U(\mu))(-[Y, \pi\sharp(\eta)]_{VF}) \rightarrow U(\mu(-[Y, \pi\sharp(\eta)]_{VF})) - \mu([U, (-[Y, \pi\sharp(\eta)]_{VF})]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad \mu(-[Y, \pi\sharp(\eta)]_{VF}) \rightarrow -\mu([Y, \pi\sharp(\eta)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& \quad [U, (-[Y, \pi\sharp(\eta)]_{VF})]_{VF} \rightarrow -[U, [Y, \pi\sharp(\eta)]_{VF}]_{VF} \quad [(-X), Y]_{VF} \rightarrow -[X, Y]_{VF} \text{ (axiom)} \\
& \quad \mu(-[U, [Y, \pi\sharp(\eta)]_{VF}]_{VF}) \rightarrow -\mu([U, [Y, \pi\sharp(\eta)]_{VF}]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp(L_U(\mu))}(\eta))(Y) \rightarrow (\pi\sharp(L_U(\mu)))(\eta(Y)) - \eta([\pi\sharp(L_U(\mu)), Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad [\pi\sharp(L_U(\mu)), Y]_{VF} \rightarrow -[Y, \pi\sharp(L_U(\mu))]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \eta(-[Y, \pi\sharp(L_U(\mu))]_{VF}) \rightarrow -\eta([Y, \pi\sharp(L_U(\mu))]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(U(-\eta(\pi\sharp(\mu)))) - \mu([U, \pi\sharp(\eta)]_{VF}))(Y) \rightarrow Y(U(-\eta(\pi\sharp(\mu)))) - \mu([U, \pi\sharp(\eta)]_{VF}) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& \quad K_{X_{\iota_U(\mu)}}(\eta) \rightarrow -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \quad [K\_V \omega = -L\_V \omega + d(\iota\_V \omega)] \text{ (axiom)} \\
& \quad K_{\pi\sharp(L_U(\mu))}(\eta) \rightarrow -L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta)) \quad [K\_V \omega = -L\_V \omega + d(\iota\_V \omega)] \text{ (axiom)} \\
& \quad K_{[U, \pi\sharp(\mu)]_{VF}}(\eta) \rightarrow -L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta)) \quad [K\_V \omega = -L\_V \omega + d(\iota\_V \omega)] \text{ (axiom)} \\
& \quad K_{X_{\iota_U(\mu)}}(\eta) \rightarrow -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \quad [K\_V \omega = -L\_V \omega + d(\iota\_V \omega)] \text{ (axiom)} \\
& \left( \left( - \left( -L_{X_{\iota_U(\eta)}}(\mu) + (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu)) \right) - L_{X_{\iota_U(\eta)}}(\mu) \right) + \left( - \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right) + ((-L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta))) - (-L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta)))) \right) \right) \\
& \left( - \left( -L_{X_{\iota_U(\eta)}}(\mu) + (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu)) \right) - L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) \rightarrow - \left( -L_{X_{\iota_U(\eta)}}(\mu) + (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu)) \right)(Y) - \left( L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right)(Y) \\
& \left( -L_{X_{\iota_U(\eta)}}(\mu) + (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu)) \right)(Y) \rightarrow - \left( L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) + (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& \left( L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) \rightarrow X_{\iota_U(\eta)}(\mu(Y)) - \mu([X_{\iota_U(\eta)}, Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad (L_{\pi\sharp(L_U(\eta))}(\mu) - L_{[U, \pi\sharp(\eta)]_{VF}}(\mu))(Y) \rightarrow (L_{\pi\sharp(L_U(\eta))}(\mu))(Y) - (L_{[U, \pi\sharp(\eta)]_{VF}}(\mu))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp(L_U(\eta))}(\mu))(Y) \rightarrow (\pi\sharp(L_U(\eta)))(\mu(Y)) - \mu([\pi\sharp(L_U(\eta)), Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad [\pi\sharp(L_U(\eta)), Y]_{VF} \rightarrow -[Y, \pi\sharp(L_U(\eta))]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \mu(-[Y, \pi\sharp(L_U(\eta))]_{VF}) \rightarrow -\mu([Y, \pi\sharp(L_U(\eta))]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{[U, \pi\sharp(\eta)]_{VF}}(\mu))(Y) \rightarrow [U, \pi\sharp(\eta)]_{VF}(\mu(Y)) - \mu([U, \pi\sharp(\eta)]_{VF}, Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad [[U, \pi\sharp(\eta)]_{VF}, Y]_{VF} \rightarrow -[Y, [U, \pi\sharp(\eta)]_{VF}]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \mu(-[Y, [U, \pi\sharp(\eta)]_{VF}]_{VF}) \rightarrow -\mu([Y, [U, \pi\sharp(\eta)]_{VF}]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& \left( L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) \rightarrow X_{\iota_U(\eta)}(\mu(Y)) - \mu([X_{\iota_U(\eta)}, Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \left( - \left( - \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right) + ((-L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta))) - (-L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta)))) \right) - \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right) \right) \\
& \left( - \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right) + ((-L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta))) - (-L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta)))) \right)(Y) \rightarrow - \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right)(Y) \\
& \left( -L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right)(Y) \rightarrow - \left( L_{X_{\iota_U(\mu)}}(\eta) \right)(Y) + \left( d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right)(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& \left( L_{X_{\iota_U(\mu)}}(\eta) \right)(Y) \rightarrow X_{\iota_U(\mu)}(\eta(Y)) - \eta([X_{\iota_U(\mu)}, Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \left( d(\iota_{X_{\iota_U(\mu)}}(\eta)) \right)(Y) \rightarrow Y \left( \iota_{X_{\iota_U(\mu)}}(\eta) \right) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& \quad Y \left( \iota_{X_{\iota_U(\mu)}}(\eta) \right) \rightarrow Y \left( \eta(X_{\iota_U(\mu)}) \right) \quad [\text{bare } \iota\_X \omega \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \quad \eta(X_{\iota_U(\mu)}) \rightarrow -(\pi\sharp(\eta))(\iota_U(\mu)) \quad [\alpha(X\_f) = -\pi\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\
& \quad (\pi\sharp(\eta))(\iota_U(\mu)) \rightarrow (\pi\sharp(\eta))(\mu(U)) \quad [\text{bare } \iota\_X \omega \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \left( (-L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta))) - (-L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta))) \right)(Y) \rightarrow (-L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta)))(Y) - (-L_{[U, \pi\sharp(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi\sharp(\mu)]_{VF}}(\eta)))(Y) \\
& \left( -L_{\pi\sharp(L_U(\mu))}(\eta) + d(\iota_{\pi\sharp(L_U(\mu))}(\eta)) \right)(Y) \rightarrow - \left( L_{\pi\sharp(L_U(\mu))}(\eta) \right)(Y) + \left( d(\iota_{\pi\sharp(L_U(\mu))}(\eta)) \right)(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp(L_U(\mu))}(\eta))(Y) \rightarrow (\pi\sharp(L_U(\mu)))(\eta(Y)) - \eta([\pi\sharp(L_U(\mu)), Y]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& \quad [\pi\sharp(L_U(\mu)), Y]_{VF} \rightarrow -[Y, \pi\sharp(L_U(\mu))]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \quad \eta(-[Y, \pi\sharp(L_U(\mu))]_{VF}) \rightarrow -\eta([Y, \pi\sharp(L_U(\mu))]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& \left( d(\iota_{\pi\sharp(L_U(\mu))}(\eta)) \right)(Y) \rightarrow Y \left( \iota_{\pi\sharp(L_U(\mu))}(\eta) \right) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& \quad Y \left( \iota_{\pi\sharp(L_U(\mu))}(\eta) \right) \rightarrow Y \left( \eta(\pi\sharp(L_U(\mu))) \right) \quad [\text{bare } \iota\_X \omega \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \quad \eta(\pi\sharp(L_U(\mu))) \rightarrow -(L_U(\mu))(\pi\sharp(\eta)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha)) \text{ } [\pi]] \text{ (axiom)}
\end{aligned}$$

$$\begin{aligned}
& (L_U(\mu))(\pi^\#(\eta)) \rightarrow U(\mu(\pi^\#(\eta))) - \mu([U, \pi^\#(\eta)]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& \mu(\pi^\#(\eta)) \rightarrow -\eta(\pi^\#(\mu)) \quad [\alpha(\pi^\#(\beta)) \rightarrow -\beta(\pi^\#(\alpha)) \text{ } [\pi]] \text{ (axiom)} \\
& (-L_{[U, \pi^\#(\mu)]_{VF}}(\eta) + d(\iota_{[U, \pi^\#(\mu)]_{VF}}(\eta)))(Y) \rightarrow -(L_{[U, \pi^\#(\mu)]_{VF}}(\eta))(Y) + (d(\iota_{[U, \pi^\#(\mu)]_{VF}}(\eta)))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{[U, \pi^\#(\mu)]_{VF}}(\eta))(Y) \rightarrow [U, \pi^\#(\mu)]_{VF}(\eta(Y)) - \eta([U, \pi^\#(\mu)]_{VF, Y}]_{VF} \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& [[U, \pi^\#(\mu)]_{VF, Y}]_{VF} \rightarrow -[Y, [U, \pi^\#(\mu)]_{VF}]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]]] \text{ (axiom)} \\
& \eta(-[Y, [U, \pi^\#(\mu)]_{VF}]_{VF}) \rightarrow -\eta([Y, [U, \pi^\#(\mu)]_{VF}]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(\iota_{[U, \pi^\#(\mu)]_{VF}}(\eta)))(Y) \rightarrow Y(\iota_{[U, \pi^\#(\mu)]_{VF}}(\eta)) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& Y(\iota_{[U, \pi^\#(\mu)]_{VF}}(\eta)) \rightarrow Y(\eta([U, \pi^\#(\mu)]_{VF})) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& (-L_{X_{\iota_U(\mu)}}(\eta) + d(\iota_{X_{\iota_U(\mu)}}(\eta)))(Y) \rightarrow -(L_{X_{\iota_U(\mu)}}(\eta))(Y) + (d(\iota_{X_{\iota_U(\mu)}}(\eta)))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{X_{\iota_U(\mu)}}(\eta))(Y) \rightarrow X_{\iota_U(\mu)}(\eta(Y)) - \eta([X_{\iota_U(\mu), Y}]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& (d(\iota_{X_{\iota_U(\mu)}}(\eta)))(Y) \rightarrow Y(\iota_{X_{\iota_U(\mu)}}(\eta)) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& Y(\iota_{X_{\iota_U(\mu)}}(\eta)) \rightarrow Y(\eta(X_{\iota_U(\mu)})) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \eta(X_{\iota_U(\mu)}) \rightarrow -(\pi^\#(\eta))(\iota_U(\mu)) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\
& (\pi^\#(\eta))(\iota_U(\mu)) \rightarrow (\pi^\#(\eta))(\mu(U)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& ((U(((\pi^\#(\eta))(\mu(Y)) - (-\mu([Y, \pi^\#(\eta)]_{VF}))) - ((\pi^\#(\mu))(\eta(Y)) - (-\eta([Y, \pi^\#(\mu)]_{VF}))) - Y(-\eta(\pi^\#(\mu)))) - ((\pi^\#(\eta))(\mu([U, Y]_{VF})) - (-\mu([U, Y]_{VF}, \pi^\#(\mu)))) \\
& (((U(((\pi^\#(\eta))(\mu(Y)) - (-U(\mu([Y, \pi^\#(\eta)]_{VF}))) - (U((\pi^\#(\mu))(\eta(Y)) - (-U(\eta([Y, \pi^\#(\mu)]_{VF})))) - (-U(Y(\eta(\pi^\#(\mu)))))) - ((\pi^\#(\eta))(\mu([U, Y]_{VF})) - (-U(Y(\eta(\pi^\#(\mu)))))) \\
& U(Y(\eta(\pi^\#(\mu)))) + U((\pi^\#(\eta))(\mu(Y)) - U((\pi^\#(\mu))(\eta(Y)) - Y(U(\eta(\pi^\#(\mu)))) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) - \eta([U, Y]_{VF}(\eta(\pi^\#(\mu)))) \\
& U(Y(\eta(\pi^\#(\mu)))) + U((\pi^\#(\eta))(\mu(Y)) - U((\pi^\#(\mu))(\eta(Y)) - Y(U(\eta(\pi^\#(\mu)))) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) + \mu([U, Y]_{VF}(\eta(\pi^\#(\mu)))) \\
& U(Y(\eta(\pi^\#(\mu)))) + U((\pi^\#(\eta))(\mu(Y)) - U((\pi^\#(\mu))(\eta(Y)) - Y(U(\eta(\pi^\#(\mu)))) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) - (\pi^\#(\eta))(U(\mu(Y))) \\
& [U, Y]_{VF}(\eta(\pi^\#(\mu))) + U((\pi^\#(\eta))(\mu(Y)) - U((\pi^\#(\mu))(\eta(Y)) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) - (\pi^\#(\eta))(U(\mu(Y))) \\
& [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, Y]_{VF}(\eta(\pi^\#(\mu))) - U((\pi^\#(\mu))(\eta(Y)) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) + (\pi^\#(\mu))(U(\eta(Y))) \\
& [\pi^\#(\mu), U]_{VF} \rightarrow -[U, \pi^\#(\mu)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]]] \text{ (axiom)} \\
& (-[U, \pi^\#(\mu)]_{VF}(\eta(Y)) + [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) \rightarrow -[U, \pi^\#(\mu)]_{VF}(\eta(Y)) \\
& -[U, \pi^\#(\mu)]_{VF}(\eta(Y)) + [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, Y]_{VF}(\eta(\pi^\#(\mu))) - [U, \pi^\#(\eta)]_{VF}(\mu(Y)) + [U, \pi^\#(\mu)]_{VF}(\eta(Y)) \rightarrow 0 \quad [\text{simplify}]
\end{aligned}$$

### 1.3.2 (2) $\mathcal{L}_{\tilde{d}\tilde{\iota}_\eta U} \mu = -[d\iota_U \eta, \mu]_K$

Sol taraftaki  $\tilde{d}\tilde{\iota}_\eta U$  bir vektör alanı ( $\tilde{\iota}_\eta$  SN-derecesini düşürür,  $\tilde{d}_\pi$  yükseltir; 1-vector  $U$ 'dan başlayınca yine 1-vector). Sağ taraftaki  $d\iota_U \eta$  bir 1-form, Koszul bracket  $\mu$  ile alınıyor.

```
[4]: head_lhs = Act(tilde_d(pi), Act(tilde_interior(eta), U))
lhs = Act(lie_derivative(head_lhs), mu)
head_rhs = Act(default_d, Act(interior(U), eta))
rhs = Neg(BracketApply(K_b, head_rhs, mu))
chain = prob.prove_derivator(lhs, rhs, eval_args=(Y,), side="form")
print(f"(2) form-side, L_{dtilde iotatilde_eta U} mu = -[d iota_U eta, mu]_K
<--> {len(chain)} adimda kapandi")
display_chain(chain)
```

(2) form-side, L\_{dtilde iotatilde\_eta U} mu = -[d iota\_U eta, mu]\_K --> 21 adimda kapandi

[4]:

$$\begin{aligned}
& (-L_{X_{\iota_U(\eta)}}(\mu))(Y) \rightarrow -(L_{X_{\iota_U(\eta)}}(\mu))(Y) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{X_{\iota_U(\eta)}}(\mu))(Y) \rightarrow X_{\iota_U(\eta)}(\mu(Y)) - \mu([X_{\iota_U(\eta), Y}]_{VF}) \quad [\text{L\_X intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& [d(\iota_U(\eta)), \mu]_{[\cdot, \cdot]_K[\pi]} \rightarrow L_{\pi^\#(d(\iota_U(\eta)))}(\mu) - L_{\pi^\#(\mu)}(d(\iota_U(\eta))) - d(\langle \pi^\#(d(\iota_U(\eta))), \mu \rangle) \quad [[\alpha, \beta]_K = L\_p\alpha(\beta) - L\_p\beta(\alpha) - d\langle \rho\alpha, \beta \rangle \text{ } [[\cdot, \cdot]_K[\pi]]] \text{ (axiom)}
\end{aligned}$$

$$\begin{aligned}
& L_{\pi\sharp(d(\iota_U(\eta)))}(\mu) \rightarrow L_{X_{\iota_U(\eta)}}(\mu) \quad [\text{atom-slot lift}] \text{ (axiom)} \\
& \pi\sharp(d(\iota_U(\eta))) \rightarrow X_{\iota_U(\eta)} \quad [\pi\sharp(\text{df}) = X\_f [\pi]] \text{ (axiom)} \\
& \langle X_{\iota_U(\eta)}, \mu \rangle \rightarrow -(\pi\sharp(\mu)(\iota_U(\eta))) \quad [\langle X\_f, \mu \rangle = -\pi\sharp(\mu)(f) [\pi]] \text{ (axiom)} \\
& (\pi\sharp(\mu)(\iota_U(\eta)) \rightarrow (\pi\sharp(\mu)(\eta(U))) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \left( - \left( L_{X_{\iota_U(\eta)}}(\mu) - L_{\pi\sharp(\mu)(d(\iota_U(\eta)))} - d(-(\pi\sharp(\mu)(\eta(U)))) \right) \right)(Y) \rightarrow \left( L_{X_{\iota_U(\eta)}}(\mu) - L_{\pi\sharp(\mu)(d(\iota_U(\eta)))} - d(-(\pi\sharp(\mu)(\eta(U)))) \right)(Y) \quad [\text{MultiEval he}] \\
& \left( L_{X_{\iota_U(\eta)}}(\mu) - L_{\pi\sharp(\mu)(d(\iota_U(\eta)))} - d(-(\pi\sharp(\mu)(\eta(U)))) \right)(Y) \rightarrow \left( L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) - \left( L_{\pi\sharp(\mu)(d(\iota_U(\eta)))}(Y) - d(-(\pi\sharp(\mu)(\eta(U))))(Y) \right) \quad [\text{MultiEval}] \\
& \left( L_{X_{\iota_U(\eta)}}(\mu) \right)(Y) \rightarrow X_{\iota_U(\eta)}(\mu(Y)) - \mu\left( [X_{\iota_U(\eta), Y}]_{VF} \right) \quad [\text{L\_X intrinsic: } (\text{L\_X } \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& (L_{\pi\sharp(\mu)(d(\iota_U(\eta)))}(Y) \rightarrow (\pi\sharp(\mu)(d(\iota_U(\eta))))(Y) - (d(\iota_U(\eta)))([\pi\sharp(\mu), Y]_{VF}) \quad [\text{L\_X intrinsic: } (\text{L\_X } \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \\
& (d(\iota_U(\eta)))(Y) \rightarrow Y(\iota_U(\eta)) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& Y(\iota_U(\eta)) \rightarrow Y(\eta(U)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& [\pi\sharp(\mu), Y]_{VF} \rightarrow -[Y, \pi\sharp(\mu)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]]] \text{ (axiom)} \\
& (d(\iota_U(\eta)))(-[Y, \pi\sharp(\mu)]_{VF}) \rightarrow (-[Y, \pi\sharp(\mu)]_{VF})(\iota_U(\eta)) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& (d(-(\pi\sharp(\mu)(\eta(U))))(Y) \rightarrow Y(-(\pi\sharp(\mu)(\eta(U)))) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& - \left( X_{\iota_U(\eta)}(\mu(Y)) - \mu\left( [X_{\iota_U(\eta), Y}]_{VF} \right) \right) - \left( - \left( \left( X_{\iota_U(\eta)}(\mu(Y)) - \mu\left( [X_{\iota_U(\eta), Y}]_{VF} \right) \right) - ((\pi\sharp(\mu))(Y(\eta(U)))) - (-[Y, \pi\sharp(\mu)]_{VF})(\iota_U(\eta)) - Y(-(\pi\sharp(\mu)(\eta(U)))) \right) \right) \\
& - \left( X_{\iota_U(\eta)}(\mu(Y)) - \mu\left( [X_{\iota_U(\eta), Y}]_{VF} \right) \right) - \left( - \left( \left( X_{\iota_U(\eta)}(\mu(Y)) - \mu\left( [X_{\iota_U(\eta), Y}]_{VF} \right) \right) - ((\pi\sharp(\mu))(Y(\eta(U)))) - (-[Y, \pi\sharp(\mu)]_{VF})(\iota_U(\eta)) - (-Y((\pi\sharp(\mu)(\eta(U)))) \right) \right) \\
& [Y, \pi\sharp(\mu)]_{VF}(\iota_U(\eta)) \rightarrow [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(\text{D}, \_) \rightarrow \text{Act}(\text{D}, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& Y((\pi\sharp(\mu)(\eta(U)))) - [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) - (\pi\sharp(\mu))(Y(\eta(U))) \rightarrow [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) - [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) \quad [X(Y(f)) - Y(X(f)) = [X, Y]_{VF}(f)] \text{ (axiom)} \\
& [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) - [Y, \pi\sharp(\mu)]_{VF}(\eta(U)) \rightarrow 0 \quad [\text{simplify}] \text{ --- canonical-form pipeline (intra-loop)}
\end{aligned}$$

### 1.3.3 (3) $0 = d \iota_{\tilde{L}_W} \eta - d \iota_{\tilde{d}_{\eta} W} \omega + d \iota_W [\omega, \eta]_K$

Sıfır-toplam özdeşliği; `prove_derivator` iki Expr alıyor, bu yüzden ilk iki terim `lhs_pair`, üçüncü (pozitif) terim de `rhs_third` olarak gruplanıp `lhs_pair = rhs_third` formatında veriliyor.

```

[5]: v1 = Act(tilde_lie(omega, pi), W)
v2 = Act(tilde_d(pi), Act(tilde_interior(eta), W))
bracket_oe = BracketApply(K_b, omega, eta)

lhs_pair = Sum(
    Act(default_d, Act(interior(v1), eta)),
    Act(default_d, Act(interior(W), bracket_oe)),
)
rhs_third = Act(default_d, Act(interior(v2), omega))
chain = prob.prove_derivator(lhs_pair, rhs_third, eval_args=(Y,), side="form")
print(f"(3) form-side, zero-sum reformulated lhs_pair = rhs_third --> ␣
    ↳{len(chain)} adimda kapandi")
display_chain(chain)

```

(3) form-side, zero-sum reformulated lhs\_pair = rhs\_third --> 30 adimda kapandi

[5]:

$$\begin{aligned}
& [\omega, \eta]_{[\cdot, \cdot]_K [\pi]} \rightarrow L_{\pi\sharp(\omega)}(\eta) - L_{\pi\sharp(\eta)}(\omega) - d(\langle \pi\sharp(\omega), \eta \rangle) \quad [[\alpha, \beta]_K = L_{\rho}\alpha(\beta) - L_{\rho}\beta(\alpha) - d\langle \rho\alpha, \beta \rangle [[\cdot, \cdot]_K [\pi]]] \text{ (axiom)} \\
& \langle \pi\sharp(\omega), \eta \rangle \rightarrow \eta(\pi\sharp(\omega)) \quad [(\alpha, X) \rightarrow \alpha(X)] \text{ (axiom)} \\
& \left( d \left( -\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta)) \right) + d(\iota_W(L_{\pi\sharp(\omega)}(\eta) - L_{\pi\sharp(\eta)}(\omega) - d(\eta(\pi\sharp(\omega)))) \right)(Y) \rightarrow \left( d \left( -\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta)) \right) \right)(Y) \\
& \left( d \left( -\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta)) \right) \right)(Y) \rightarrow Y \left( -\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta)) \right) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \\
& (d(\iota_W(L_{\pi\sharp(\omega)}(\eta) - L_{\pi\sharp(\eta)}(\omega) - d(\eta(\pi\sharp(\omega))))(Y) \rightarrow Y(\iota_W(L_{\pi\sharp(\omega)}(\eta) - L_{\pi\sharp(\eta)}(\omega) - d(\eta(\pi\sharp(\omega)))) \quad [\text{d intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \text{hat\_i}, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)]
\end{aligned}$$

$$\begin{aligned}
& Y(\iota_W(L_{\pi\sharp}(\omega)(\eta) - L_{\pi\sharp}(\eta)(\omega) - d(\eta(\pi\sharp(\omega)))) \rightarrow Y((L_{\pi\sharp}(\omega)(\eta) - L_{\pi\sharp}(\eta)(\omega) - d(\eta(\pi\sharp(\omega))))(W)) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& (L_{\pi\sharp}(\omega)(\eta) - L_{\pi\sharp}(\eta)(\omega) - d(\eta(\pi\sharp(\omega))))(W) \rightarrow (L_{\pi\sharp}(\omega)(\eta))(W) - (L_{\pi\sharp}(\eta)(\omega))(W) - (d(\eta(\pi\sharp(\omega))))(W) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\omega)(\eta))(W) \rightarrow (\pi\sharp(\omega))(\eta(W)) - \eta([\pi\sharp(\omega), W]_{VF}) \quad [\text{L}_X \text{ intrinsic: } (L_X \omega)(Y_1, \dots) = X(\omega(Y_1, \dots)) - \Sigma \omega(\dots, [X, Y_i]_{VF}, \dots)] \text{ (axiom)} \\
& [\pi\sharp(\omega), W]_{VF} \rightarrow -[W, \pi\sharp(\omega)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \eta(-[W, \pi\sharp(\omega)]_{VF}) \rightarrow -\eta([W, \pi\sharp(\omega)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (L_{\pi\sharp}(\eta)(\omega))(W) \rightarrow (\pi\sharp(\eta))(\omega(W)) - \omega([\pi\sharp(\eta), W]_{VF}) \quad [\text{L}_X \text{ intrinsic: } (L_X \omega)(Y_1, \dots) = X(\omega(Y_1, \dots)) - \Sigma \omega(\dots, [X, Y_i]_{VF}, \dots)] \text{ (axiom)} \\
& [\pi\sharp(\eta), W]_{VF} \rightarrow -[W, \pi\sharp(\eta)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \omega(-[W, \pi\sharp(\eta)]_{VF}) \rightarrow -\omega([W, \pi\sharp(\eta)]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (d(\eta(\pi\sharp(\omega))))(W) \rightarrow W(\eta(\pi\sharp(\omega))) \quad [\text{d intrinsic (Koszul): } (d\omega)(X_0, \dots, X_p) = \Sigma \pm X_i(\omega(\dots, \text{hat}_i, \dots)) + \Sigma \pm \omega([X_i, X_j]_{VF}, \dots)] \text{ (axiom)} \\
& (d(-\iota_{X_{\iota_W(\omega)}}(\omega)))(Y) \rightarrow Y(-\iota_{X_{\iota_W(\omega)}}(\omega)) \quad [\text{d intrinsic (Koszul): } (d\omega)(X_0, \dots, X_p) = \Sigma \pm X_i(\omega(\dots, \text{hat}_i, \dots)) + \Sigma \pm \omega([X_i, X_j]_{VF}, \dots)] \text{ (axiom)} \\
& (Y(-\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta))) + Y(((\pi\sharp(\omega))(\eta(W)) - (\eta([W, \pi\sharp(\omega)]_{VF}))) - ((\pi\sharp(\eta))(\omega(W)) - (-\omega([W, \pi\sharp(\eta)]_{VF}))) - W \\
& ((-Y(\iota_{X_{\iota_W(\omega)}}(\eta) + (\iota_{\pi\sharp(L_W(\omega))}(\eta) - \iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta))) + Y((\pi\sharp(\omega))(\eta(W))) - (-Y(\eta([W, \pi\sharp(\omega)]_{VF}))) - (Y((\pi\sharp(\eta))(\omega(W))) - (-Y(\omega([W, \pi\sharp(\eta)]_{VF}))) - W \\
& Y(\iota_{X_{\iota_W(\omega)}}(\omega)) \rightarrow Y(\omega(X_{\iota_W(\omega)})) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \omega(X_{\iota_W(\omega)}) \rightarrow -(\pi\sharp(\omega))(\iota_W(\eta)) \quad [\alpha(X_f) = -\pi\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\
& (\pi\sharp(\omega))(\iota_W(\eta)) \rightarrow (\pi\sharp(\omega))(\eta(W)) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& Y(\iota_{X_{\iota_W(\omega)}}(\eta)) \rightarrow Y(\eta(X_{\iota_W(\omega)})) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \eta(X_{\iota_W(\omega)}) \rightarrow -(\pi\sharp(\eta))(\iota_W(\omega)) \quad [\alpha(X_f) = -\pi\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\
& (\pi\sharp(\eta))(\iota_W(\omega)) \rightarrow (\pi\sharp(\eta))(\omega(W)) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& Y(\iota_{[W, \pi\sharp(\omega)]_{VF}}(\eta)) \rightarrow Y(\eta([W, \pi\sharp(\omega)]_{VF})) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& Y(\iota_{\pi\sharp(L_W(\omega))}(\eta)) \rightarrow Y(\eta(\pi\sharp(L_W(\omega)))) \quad [\text{bare } \iota_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ (axiom)} \\
& \eta(\pi\sharp(L_W(\omega))) \rightarrow -(L_W(\omega))(\pi\sharp(\eta)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha)) \text{ } [\pi]] \text{ (axiom)} \\
& (L_W(\omega))(\pi\sharp(\eta)) \rightarrow W(\omega(\pi\sharp(\eta))) - \omega([W, \pi\sharp(\eta)]_{VF}) \quad [\text{L}_X \text{ intrinsic: } (L_X \omega)(Y_1, \dots) = X(\omega(Y_1, \dots)) - \Sigma \omega(\dots, [X, Y_i]_{VF}, \dots)] \text{ (axiom)} \\
& \omega(\pi\sharp(\eta)) \rightarrow -\eta(\pi\sharp(\omega)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha)) \text{ } [\pi]] \text{ (axiom)} \\
& -Y(W(\eta(\pi\sharp(\omega)))) + Y(\eta([W, \pi\sharp(\omega)]_{VF})) + Y(-(\pi\sharp(\omega))(\eta(W))) - Y(-(\pi\sharp(\eta))(\omega(W))) - Y(\eta([W, \pi\sharp(\omega)]_{VF})) + Y(-(W(-\eta(\pi\sharp(\omega)))) - \omega([W, \pi\sharp(\eta)]_{VF})) \\
& -Y(W(\eta(\pi\sharp(\omega)))) + Y(\eta([W, \pi\sharp(\omega)]_{VF})) - Y((\pi\sharp(\omega))(\eta(W))) - (-Y((\pi\sharp(\eta))(\omega(W)))) - Y(\eta([W, \pi\sharp(\omega)]_{VF})) - (-Y(W(\eta(\pi\sharp(\omega)))) - Y(\omega([W, \pi\sharp(\eta)]_{VF}))
\end{aligned}$$

## 1.4 3. Dual multivector tarafi

$$1.4.1 \quad (1') \quad \tilde{D}_{\tilde{\mathcal{L}}_\eta}^{\text{SN}}(U, V) = \tilde{\mathcal{L}}_{K_U \eta} V + \tilde{K}_{K_V \eta} U$$

Sol taraftaki  $\tilde{D}_{\tilde{\mathcal{L}}_\eta}^{\text{SN}}(U, V) = \tilde{\mathcal{L}}_\eta[U, V]_{\text{SN}} - [\tilde{\mathcal{L}}_\eta U, V]_{\text{SN}} - [U, \tilde{\mathcal{L}}_\eta V]_{\text{SN}}$  obstruction'ı; sağ tarafta  $K$  (form Cartan-remainder) ile bağlanan iki Lichnerowicz düzeltmesi. Engine 117 adımda kapatıyor — Lichnerowicz açılımı + SN/LBVF Sum-doğrusallığı + Sharp-Neg-doğrusallığı + LBVF Jacobi zinciri üzerinden.

```
[6]: lhs = derivator(tilde_lie(eta, pi), default_sn, U, V)
K_U_eta = Act(K(U), eta)
K_V_eta = Act(K(V), eta)
rhs = Sum(
    Act(tilde_lie(K_U_eta, pi), V),
    Act(K_tilde(K_V_eta, pi), U),
)
chain = prob.prove_derivator(lhs, rhs, eval_args=(xi,), side="multivector")
print(f"(1') multivec, Dtilde^SN_{{Ltilde_eta}}(U,V) = Ltilde_{{K_U_eta}} V +_L
    Ktilde_{{K_V_eta}} U --> {len(chain)} adimda kapandi")
display_chain(chain)
```



(1') multivec,  $\text{Dtilde}^{\wedge}\text{SN}_{\{\text{Ltilde}_{\eta}\}}(\text{U}, \text{V}) = \text{Ltilde}_{\{\text{K}_U \eta\}} \text{V} + \text{Ktilde}_{\{\text{K}_V \eta\}} \text{U} \rightarrow 117 \text{ adimda kapandi}$

[6]:

$$\begin{aligned}
& [U, V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [U, V]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{\eta}([U, V]_{VF}) \rightarrow -X_{\eta}([U, V]_{VF}) + \pi\sharp(L_{[U, V]_{VF}}(\eta)) - [[U, V]_{VF, \pi\sharp(\eta)}]_{VF} \quad [\tilde{L}_{\alpha}(U) = -X_{\alpha}(U) + \pi\sharp(L_{\alpha} U) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{\eta}(U) \rightarrow -X_{\eta}(U) + \pi\sharp(L_U(\eta)) - [U, \pi\sharp(\eta)]_{VF} \quad [\tilde{L}_{\alpha}(U) = -X_{\alpha}(U) + \pi\sharp(L_{\alpha} U) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& [-X_{\eta}(U) + \pi\sharp(L_U(\eta)) - [U, \pi\sharp(\eta)]_{VF}, V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [-X_{\eta}(U), V]_{[\cdot, \cdot]_{\text{SN}}} + [\pi\sharp(L_U(\eta)), V]_{[\cdot, \cdot]_{\text{SN}}} + [-[U, \pi\sharp(\eta)]_{VF}, V]_{[\cdot, \cdot]_{\text{SN}}} \quad [[\text{Sum}(a, b, \dots), Z]_{\text{SN}}] \\
& [-X_{\eta}(U), V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow -[X_{\eta}(U), V]_{[\cdot, \cdot]_{\text{SN}}} \quad [(-a), b]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} / [a, (-b)]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} \text{ (axiom)} \\
& [X_{\eta}(U), V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [X_{\eta(U), V}]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& [X_{\eta(U), V}]_{VF} \rightarrow -[V, X_{\eta(U)}]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& [\pi\sharp(L_U(\eta)), V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [\pi\sharp(L_U(\eta)), V]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& [\pi\sharp(L_U(\eta)), V]_{VF} \rightarrow -[V, \pi\sharp(L_U(\eta))]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& [-[U, \pi\sharp(\eta)]_{VF}, V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow -[[U, \pi\sharp(\eta)]_{VF}, V]_{[\cdot, \cdot]_{\text{SN}}} \quad [(-a), b]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} / [a, (-b)]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} \text{ (axiom)} \\
& [[U, \pi\sharp(\eta)]_{VF}, V]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [[U, \pi\sharp(\eta)]_{VF, V}]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& [[U, \pi\sharp(\eta)]_{VF, V}]_{VF} \rightarrow -[V, [U, \pi\sharp(\eta)]_{VF}]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{\eta}(V) \rightarrow -X_{\eta}(V) + \pi\sharp(L_V(\eta)) - [V, \pi\sharp(\eta)]_{VF} \quad [\tilde{L}_{\alpha}(U) = -X_{\alpha}(U) + \pi\sharp(L_{\alpha} U) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& [U, -X_{\eta}(V) + \pi\sharp(L_V(\eta)) - [V, \pi\sharp(\eta)]_{VF}]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [U, -X_{\eta}(V)]_{[\cdot, \cdot]_{\text{SN}}} + [U, \pi\sharp(L_V(\eta))]_{[\cdot, \cdot]_{\text{SN}}} + [U, -[V, \pi\sharp(\eta)]_{VF}]_{[\cdot, \cdot]_{\text{SN}}} \quad [[\text{Sum}(a, b, \dots), Z]_{\text{SN}}] \\
& [U, -X_{\eta}(V)]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow -[U, X_{\eta}(V)]_{[\cdot, \cdot]_{\text{SN}}} \quad [(-a), b]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} / [a, (-b)]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} \text{ (axiom)} \\
& [U, X_{\eta}(V)]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [U, X_{\eta(V)}]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& [U, \pi\sharp(L_V(\eta))]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [U, \pi\sharp(L_V(\eta))]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& [U, -[V, \pi\sharp(\eta)]_{VF}]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow -[U, [V, \pi\sharp(\eta)]_{VF}]_{[\cdot, \cdot]_{\text{SN}}} \quad [(-a), b]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} / [a, (-b)]_{\text{SN}} \rightarrow -[a, b]_{\text{SN}} \text{ (axiom)} \\
& [U, [V, \pi\sharp(\eta)]_{VF}]_{[\cdot, \cdot]_{\text{SN}}} \rightarrow [U, [V, \pi\sharp(\eta)]_{VF, V}]_{VF} \quad [[U, V]_{\text{SN}} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
& ((-X_{\eta}([U, V]_{VF}) + \pi\sharp(L_{[U, V]_{VF}}(\eta)) - [[U, V]_{VF, \pi\sharp(\eta)}]_{VF}) - (-([V, X_{\eta(U)}]_{VF}) - [V, \pi\sharp(L_U(\eta))]_{VF}) - (-[V, [U, \pi\sharp(\eta)]_{VF}]_{VF})) - (-[U, X_{\eta(V)}]_{VF} \\
& (-X_{\eta}([U, V]_{VF}) + \pi\sharp(L_{[U, V]_{VF}}(\eta)) - [[U, V]_{VF, \pi\sharp(\eta)}]_{VF})(\xi) \rightarrow -X_{\eta}([U, V]_{VF})(\xi) + (\pi\sharp(L_{[U, V]_{VF}}(\eta)))(\xi) - [[U, V]_{VF, \pi\sharp(\eta)}]_{VF}(\xi) \quad [\text{MultiEval head}] \\
& X_{\eta}([U, V]_{VF})(\xi) \rightarrow \xi(X_{\eta}([U, V]_{VF})) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(X_{\eta}([U, V]_{VF})) \rightarrow -(\pi\sharp(\xi))(\eta([U, V]_{VF})) \quad [\alpha(X_{\alpha}) = -\pi\sharp(\alpha)(f) [\pi]] \text{ (axiom)} \\
& (\pi\sharp(L_{[U, V]_{VF}}(\eta)))(\xi) \rightarrow \xi(\pi\sharp(L_{[U, V]_{VF}}(\eta))) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(\pi\sharp(L_{[U, V]_{VF}}(\eta))) \rightarrow -(L_{[U, V]_{VF}}(\eta))(\pi\sharp(\xi)) \quad [\alpha(\pi\sharp(\beta)) \rightarrow -\beta(\pi\sharp(\alpha)) [\pi]] \text{ (axiom)} \\
& (L_{[U, V]_{VF}}(\eta))(\pi\sharp(\xi)) \rightarrow [U, V]_{VF}(\eta(\pi\sharp(\xi))) - \eta([U, V]_{VF, \pi\sharp(\xi)})_{VF} \quad [L_{\alpha} X \text{ intrinsic: } (L_{\alpha} X)(Y_1, \dots) = X(\omega(Y_1, \dots)) - \Sigma \omega(\dots, [X, Y_1, \dots])] \\
& [[U, V]_{VF, \pi\sharp(\eta)}]_{VF}(\xi) \rightarrow \xi([U, V]_{VF, \pi\sharp(\eta)})_{VF} \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (-([V, X_{\eta(U)}]_{VF}) - [V, \pi\sharp(L_U(\eta))]_{VF} - (-[V, [U, \pi\sharp(\eta)]_{VF}]_{VF})(\xi) \rightarrow -([V, X_{\eta(U)}]_{VF})(\xi) - [V, \pi\sharp(L_U(\eta))]_{VF}(\xi) - (-[V, [U, \pi\sharp(\eta)]_{VF}]_{VF})(\xi) \\
& (-[V, X_{\eta(U)}]_{VF})(\xi) \rightarrow -[V, X_{\eta(U)}]_{VF}(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& [V, X_{\eta(U)}]_{VF}(\xi) \rightarrow \xi([V, X_{\eta(U)}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& [V, \pi\sharp(L_U(\eta))]_{VF}(\xi) \rightarrow \xi([V, \pi\sharp(L_U(\eta))]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (-[V, [U, \pi\sharp(\eta)]_{VF}]_{VF})(\xi) \rightarrow -[V, [U, \pi\sharp(\eta)]_{VF}]_{VF}(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& [V, [U, \pi\sharp(\eta)]_{VF}]_{VF}(\xi) \rightarrow \xi([V, [U, \pi\sharp(\eta)]_{VF}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (-[U, X_{\eta(V)}]_{VF} + [U, \pi\sharp(L_V(\eta))]_{VF} - [U, [V, \pi\sharp(\eta)]_{VF}]_{VF})(\xi) \rightarrow -[U, X_{\eta(V)}]_{VF}(\xi) + [U, \pi\sharp(L_V(\eta))]_{VF}(\xi) - [U, [V, \pi\sharp(\eta)]_{VF}]_{VF}(\xi) \quad [\text{MultiEval head}] \\
& [U, X_{\eta(V)}]_{VF}(\xi) \rightarrow \xi([U, X_{\eta(V)}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& [U, \pi\sharp(L_V(\eta))]_{VF}(\xi) \rightarrow \xi([U, \pi\sharp(L_V(\eta))]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& [U, [V, \pi\sharp(\eta)]_{VF}]_{VF}(\xi) \rightarrow \xi([U, [V, \pi\sharp(\eta)]_{VF}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{L_U(\eta)}(V) \rightarrow -X_{L_U(\eta)}(V) + \pi\sharp(L_V(L_U(\eta))) - [V, \pi\sharp(L_U(\eta))]_{VF} \quad [\tilde{L}_{\alpha}(U) = -X_{\alpha}(U) + \pi\sharp(L_{\alpha} U) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{d(L_U(\eta))}(V) \rightarrow -X_{d(L_U(\eta))}(V) + \pi\sharp(L_V(d(L_U(\eta)))) - [V, \pi\sharp(d(L_U(\eta)))]_{VF} \quad [\tilde{L}_{\alpha}(U) = -X_{\alpha}(U) + \pi\sharp(L_{\alpha} U) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& L_V(d(L_U(\eta))) \rightarrow d(L_V(L_U(\eta))) \quad [L_{\alpha} X \circ d = d \circ L_{\alpha} X \text{ (tilde, any mode)}] \text{ (axiom)} \\
& \pi\sharp(d(L_V(L_U(\eta)))) \rightarrow X_{L_V(L_U(\eta))} \quad [\pi\sharp(df) = X_{\alpha} f [\pi]] \text{ (axiom)}
\end{aligned}$$

$$\begin{aligned}
& [V, \pi^\#(d(\iota_U(\eta)))_{VF}] \rightarrow [V, X_{\eta(U)}]_{VF} \quad [\text{bare-atom-slot lift}] \text{ (axiom)} \\
& \tilde{K}_{L_V(\eta)}(U) \rightarrow -\tilde{\mathcal{L}}_{L_V(\eta)}(U) + \tilde{d}(\tilde{\iota}_{L_V(\eta)}(U)) \quad [\tilde{K}_-\eta V = -\tilde{L}_-\eta V + \tilde{d}(\tilde{\iota}_-\eta V)] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{L_V(\eta)}(U) \rightarrow -X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF} \quad [\tilde{L}_-\alpha(U) = -X_-\alpha(U) + \pi^\#(L_-U \alpha) - [U, \pi^\# \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& \tilde{d}(\tilde{\iota}_{L_V(\eta)}(U)) \rightarrow \tilde{d}(U(L_V(\eta))) \quad [\text{bare } \tilde{\iota}_-\omega(V) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(V, \omega)) (V \text{ deg } 1)] \text{ (axiom)} \\
& U(L_V(\eta)) \rightarrow (L_V(\eta))(U) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (L_V(\eta))(U) \rightarrow V(\eta(U)) - \eta([V, U]_{VF}) \quad [L_-X \text{ intrinsic: } (L_-X \omega)(Y_-1, \dots) = X(\omega(Y_-1, \dots)) - \Sigma \omega(\dots, [X, Y_-i]_{VF}, \dots)] \text{ (axiom)} \\
& [V, U]_{VF} \rightarrow -[U, V]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
& \eta(-[U, V]_{VF}) \rightarrow -\eta([U, V]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& \tilde{d}(V(\eta(U)) - (-\eta([U, V]_{VF}))) \rightarrow -X_{(V(\eta(U)) + (-\eta([U, V]_{VF})))} \quad [\text{Act}(\tilde{d}_-\pi, f) \rightarrow -X_-f \text{ [geometer sign]}] \text{ (axiom)} \\
& \tilde{K}_{d(\iota_V(\eta))}(U) \rightarrow -\tilde{\mathcal{L}}_{d(\iota_V(\eta))}(U) + \tilde{d}(\tilde{\iota}_{d(\iota_V(\eta))}(U)) \quad [\tilde{K}_-\eta V = -\tilde{L}_-\eta V + \tilde{d}(\tilde{\iota}_-\eta V)] \text{ (axiom)} \\
& \tilde{\mathcal{L}}_{d(\iota_V(\eta))}(U) \rightarrow -X_{d(\iota_V(\eta))}(U) + \pi^\#(L_U(d(\iota_V(\eta)))) - [U, \pi^\#(d(\iota_V(\eta)))]_{VF} \quad [\tilde{L}_-\alpha(U) = -X_-\alpha(U) + \pi^\#(L_-U \alpha) - [U, \pi^\# \alpha]_{VF} [\pi]] \text{ (axiom)} \\
& L_U(d(\iota_V(\eta))) \rightarrow d(L_U(\iota_V(\eta))) \quad [L_-X \circ d = d \circ L_-X \text{ (tilde, any mode)}] \text{ (axiom)} \\
& \pi^\#(d(L_U(\iota_V(\eta)))) \rightarrow X_{L_U(\iota_V(\eta))} \quad [\pi^\#(df) = X_-f [\pi]] \text{ (axiom)} \\
& [U, \pi^\#(d(\iota_V(\eta)))]_{VF} \rightarrow [U, X_{\eta(V)}]_{VF} \quad [\text{bare-atom-slot lift}] \text{ (axiom)} \\
& \tilde{d}(\tilde{\iota}_{d(\iota_V(\eta))}(U)) \rightarrow \tilde{d}(U(d(\iota_V(\eta)))) \quad [\text{bare } \tilde{\iota}_-\omega(V) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(V, \omega)) (V \text{ deg } 1)] \text{ (axiom)} \\
& U(d(\iota_V(\eta))) \rightarrow (d(\iota_V(\eta)))(U) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (d(\iota_V(\eta)))(U) \rightarrow U(\iota_V(\eta)) \quad [d \text{ intrinsic (Koszul): } (d\omega)(X_-0, \dots, X_-p) = \Sigma \pm X_-i(\omega(\dots, \text{hat}_i, \dots)) + \Sigma \pm \omega([X_-i, X_-j]_{VF}, \dots)] \text{ (axiom)} \\
& U(\iota_V(\eta)) \rightarrow U(\eta(V)) \quad [\text{bare } \iota_-X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]}] \text{ (axiom)} \\
& \tilde{d}(U(\eta(V))) \rightarrow -X_U(\eta(V)) \quad [\text{Act}(\tilde{d}_-\pi, f) \rightarrow -X_-f \text{ [geometer sign]}] \text{ (axiom)} \\
& \left( \left( -(-X_{L_U(\eta)}(V) + \pi^\#(L_V(L_U(\eta))) - [V, \pi^\#(L_U(\eta))]_{VF}) + (-X_{d(\iota_U(\eta))}(V) + X_{L_V(\iota_U(\eta))} - [V, X_{\eta(U)}]_{VF}) \right) + \left( -(-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) + (-X_{d(\iota_V(\eta))}(U) + X_{L_U(\iota_V(\eta))} - [U, X_{\eta(V)}]_{VF}) \right) \right) (\xi) \rightarrow -(-X_{L_U(\eta)}(V) + \pi^\#(L_V(L_U(\eta))) - [V, \pi^\#(L_U(\eta))]_{VF}) (\xi) \rightarrow -(-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) (\xi) \quad [\text{MultiEval head linearity}] \\
& (-X_{L_U(\eta)}(V) + \pi^\#(L_V(L_U(\eta))) - [V, \pi^\#(L_U(\eta))]_{VF}) (\xi) \rightarrow -X_{L_U(\eta)}(V) (\xi) + (\pi^\#(L_V(L_U(\eta)))) (\xi) - [V, \pi^\#(L_U(\eta))]_{VF} (\xi) \quad [\text{MultiEval head linearity}] \\
& X_{L_U(\eta)}(V) (\xi) \rightarrow \xi(X_{L_U(\eta)}(V)) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(X_{L_U(\eta)}(V)) \rightarrow -(\pi^\#(\xi))((L_U(\eta))(V)) \quad [\alpha(X_-f) = -\pi^\#(\alpha)(f) [\pi]] \text{ (axiom)} \\
& (L_U(\eta))(V) \rightarrow U(\eta(V)) - \eta([U, V]_{VF}) \quad [L_-X \text{ intrinsic: } (L_-X \omega)(Y_-1, \dots) = X(\omega(Y_-1, \dots)) - \Sigma \omega(\dots, [X, Y_-i]_{VF}, \dots)] \text{ (axiom)} \\
& (\pi^\#(L_V(L_U(\eta)))) (\xi) \rightarrow \xi(\pi^\#(L_V(L_U(\eta)))) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(\pi^\#(L_V(L_U(\eta)))) \rightarrow -(L_V(L_U(\eta)))(\pi^\#(\xi)) \quad [\alpha(\pi^\#(\beta)) \rightarrow -\beta(\pi^\#(\alpha)) [\pi]] \text{ (axiom)} \\
& (L_V(L_U(\eta)))(\pi^\#(\xi)) \rightarrow V((L_U(\eta))(\pi^\#(\xi))) - (L_U(\eta))([V, \pi^\#(\xi)]_{VF}) \quad [L_-X \text{ intrinsic: } (L_-X \omega)(Y_-1, \dots) = X(\omega(Y_-1, \dots)) - \Sigma \omega(\dots, [X, Y_-i]_{VF}, \dots)] \text{ (axiom)} \\
& (L_U(\eta))(\pi^\#(\xi)) \rightarrow U(\eta(\pi^\#(\xi))) - \eta([U, \pi^\#(\xi)]_{VF}) \quad [L_-X \text{ intrinsic: } (L_-X \omega)(Y_-1, \dots) = X(\omega(Y_-1, \dots)) - \Sigma \omega(\dots, [X, Y_-i]_{VF}, \dots)] \text{ (axiom)} \\
& (L_U(\eta))([V, \pi^\#(\xi)]_{VF}) \rightarrow U(\eta([V, \pi^\#(\xi)]_{VF})) - \eta([U, [V, \pi^\#(\xi)]_{VF}]_{VF}) \quad [L_-X \text{ intrinsic: } (L_-X \omega)(Y_-1, \dots) = X(\omega(Y_-1, \dots)) - \Sigma \omega(\dots, [X, Y_-i]_{VF}, \dots)] \text{ (axiom)} \\
& [V, \pi^\#(L_U(\eta))]_{VF} (\xi) \rightarrow \xi([V, \pi^\#(L_U(\eta))]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& (-X_{d(\iota_U(\eta))}(V) + X_{L_V(\iota_U(\eta))} - [V, X_{\eta(U)}]_{VF}) (\xi) \rightarrow -X_{d(\iota_U(\eta))}(V) (\xi) + X_{L_V(\iota_U(\eta))} (\xi) - [V, X_{\eta(U)}]_{VF} (\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& X_{d(\iota_U(\eta))}(V) (\xi) \rightarrow \xi(X_{d(\iota_U(\eta))}(V)) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(X_{d(\iota_U(\eta))}(V)) \rightarrow -(\pi^\#(\xi))((d(\iota_U(\eta)))(V)) \quad [\alpha(X_-f) = -\pi^\#(\alpha)(f) [\pi]] \text{ (axiom)} \\
& (d(\iota_U(\eta)))(V) \rightarrow V(\iota_U(\eta)) \quad [d \text{ intrinsic (Koszul): } (d\omega)(X_-0, \dots, X_-p) = \Sigma \pm X_-i(\omega(\dots, \text{hat}_i, \dots)) + \Sigma \pm \omega([X_-i, X_-j]_{VF}, \dots)] \text{ (axiom)} \\
& V(\iota_U(\eta)) \rightarrow V(\eta(U)) \quad [\text{bare } \iota_-X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]}] \text{ (axiom)} \\
& X_{L_V(\iota_U(\eta))} (\xi) \rightarrow \xi(X_{L_V(\iota_U(\eta))}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \xi(X_{L_V(\iota_U(\eta))}) \rightarrow -(\pi^\#(\xi))(L_V(\iota_U(\eta))) \quad [\alpha(X_-f) = -\pi^\#(\alpha)(f) [\pi]] \text{ (axiom)} \\
& (\pi^\#(\xi))(L_V(\iota_U(\eta))) \rightarrow (\pi^\#(\xi))(V(\iota_U(\eta))) \quad [\text{Act}(D, L_-X(f)) \rightarrow \text{Act}(D, X(f)) \text{ [D vf-like, f 0-form]}] \text{ (axiom)} \\
& V(\iota_U(\eta)) \rightarrow V(\eta(U)) \quad [\text{bare } \iota_-X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]}] \text{ (axiom)} \\
& [V, X_{\eta(U)}]_{VF} (\xi) \rightarrow \xi([V, X_{\eta(U)}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
& \left( -(-(-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) - X_{(V(\eta(U)) + (-\eta([U, V]_{VF}))))} + \left( -(-X_{d(\iota_V(\eta))}(U) + X_{L_U(\iota_V(\eta))} - [U, X_{\eta(V)}]_{VF}) \right) \right) (\xi) \rightarrow -(-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) (\xi) \rightarrow -(-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) (\xi) \quad [\text{MultiEval head linearity}] \\
& (-X_{L_V(\eta)}(U) + \pi^\#(L_U(L_V(\eta))) - [U, \pi^\#(L_V(\eta))]_{VF}) (\xi) \rightarrow -X_{L_V(\eta)}(U) (\xi) + (\pi^\#(L_U(L_V(\eta)))) (\xi) - [U, \pi^\#(L_V(\eta))]_{VF} (\xi) \quad [\text{MultiEval head linearity}] \\
& X_{L_V(\eta)}(U) (\xi) \rightarrow \xi(X_{L_V(\eta)}(U)) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)}
\end{aligned}$$

$$\begin{aligned}
& \xi(X_{L_V(\eta)(U)}) \rightarrow -(\pi^\#(\xi))(L_V(\eta)(U)) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \ [\pi]] \text{ (axiom)} \\
& (L_V(\eta))(U) \rightarrow V(\eta(U)) - \eta([V, U]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& [V, U]_{VF} \rightarrow -[U, V]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]]] \text{ (axiom)} \\
& \eta(-[U, V]_{VF}) \rightarrow -\eta([U, V]_{VF}) \quad [\text{MultiEval arg slot linearity}] \text{ (axiom)} \\
& (\pi^\#(L_U(L_V(\eta))))(\xi) \rightarrow \xi(\pi^\#(L_U(L_V(\eta)))) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& \xi(\pi^\#(L_U(L_V(\eta)))) \rightarrow -(L_U(L_V(\eta)))(\pi^\#(\xi)) \quad [\alpha(\pi^\#(\beta)) \rightarrow -\beta(\pi^\#(\alpha)) \ [\pi]] \text{ (axiom)} \\
& (L_U(L_V(\eta)))(\pi^\#(\xi)) \rightarrow U((L_V(\eta))(\pi^\#(\xi))) - (L_V(\eta))([U, \pi^\#(\xi)]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& (L_V(\eta))(\pi^\#(\xi)) \rightarrow V(\eta(\pi^\#(\xi))) - \eta([V, \pi^\#(\xi)]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& (L_V(\eta))([U, \pi^\#(\xi)]_{VF}) \rightarrow V(\eta([U, \pi^\#(\xi)]_{VF})) - \eta([V, [U, \pi^\#(\xi)]_{VF}]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\
& [U, \pi^\#(L_V(\eta))]_{VF}(\xi) \rightarrow \xi([U, \pi^\#(L_V(\eta))]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& X_{(V(\eta(U)) + (-\eta([U, V]_{VF})))}(\xi) \rightarrow \xi(X_{(V(\eta(U)) + (-\eta([U, V]_{VF})))}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& \xi(X_{(V(\eta(U)) + (-\eta([U, V]_{VF})))}) \rightarrow -(\pi^\#(\xi))(V(\eta(U)) - (-\eta([U, V]_{VF}))) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \ [\pi]] \text{ (axiom)} \\
& (-(-X_{d(\iota_V(\eta))(U)} + X_{L_{U(\iota_V(\eta))}} - [U, X_{\eta(V)}]_{VF}) - X_U(\eta(V)))(\xi) \rightarrow -(-X_{d(\iota_V(\eta))(U)} + X_{L_{U(\iota_V(\eta))}} - [U, X_{\eta(V)}]_{VF})(\xi) - X_U(\eta(V))(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& (-X_{d(\iota_V(\eta))(U)} + X_{L_{U(\iota_V(\eta))}} - [U, X_{\eta(V)}]_{VF})(\xi) \rightarrow -X_{d(\iota_V(\eta))(U)}(\xi) + X_{L_{U(\iota_V(\eta))}}(\xi) - [U, X_{\eta(V)}]_{VF}(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
& X_{d(\iota_V(\eta))(U)}(\xi) \rightarrow \xi(X_{d(\iota_V(\eta))(U)}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& \xi(X_{d(\iota_V(\eta))(U)}) \rightarrow -(\pi^\#(\xi))(d(\iota_V(\eta))(U)) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \ [\pi]] \text{ (axiom)} \\
& (d(\iota_V(\eta))(U) \rightarrow U(\iota_V(\eta)) \quad [d \text{ intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i \omega(\dots, \hat{\omega}_i, \dots) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\
& U(\iota_V(\eta)) \rightarrow U(\eta(V)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]]] \text{ (axiom)} \\
& X_{L_{U(\iota_V(\eta))}}(\xi) \rightarrow \xi(X_{L_{U(\iota_V(\eta))}}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& \xi(X_{L_{U(\iota_V(\eta))}}) \rightarrow -(\pi^\#(\xi))(L_U(\iota_V(\eta))) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \ [\pi]] \text{ (axiom)} \\
& (\pi^\#(\xi))(L_U(\iota_V(\eta))) \rightarrow (\pi^\#(\xi))(U(\iota_V(\eta))) \quad [\text{Act}(D, L\_X(f)) \rightarrow \text{Act}(D, X(f)) \text{ [D vf-like, f 0-form]]] \text{ (axiom)} \\
& U(\iota_V(\eta)) \rightarrow U(\eta(V)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]]] \text{ (axiom)} \\
& [U, X_{\eta(V)}]_{VF}(\xi) \rightarrow \xi([U, X_{\eta(V)}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& X_U(\eta(V))(\xi) \rightarrow \xi(X_U(\eta(V))) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]]] \text{ (axiom)} \\
& \xi(X_U(\eta(V))) \rightarrow -(\pi^\#(\xi))(U(\eta(V))) \quad [\alpha(X\_f) = -\pi^\#(\alpha)(f) \ [\pi]] \text{ (axiom)} \\
& ((-(-(\pi^\#(\xi))(\eta([U, V]_{VF}))) - ([U, V]_{VF}(\eta(\pi^\#(\xi)))) - \eta([U, V]_{VF}, \pi^\#(\xi)]_{VF})) - \xi([U, V]_{VF}, \pi^\#(\eta)]_{VF})) - (-(-\xi([V, X_{\eta(U)}]_{VF})) - \xi([V, \pi^\#(L_U(\eta))]_{VF})) \\
& ((-(-(\pi^\#(\xi))(\eta([U, V]_{VF}))) - ([U, V]_{VF}(\eta(\pi^\#(\xi)))) - \eta([U, V]_{VF}, \pi^\#(\xi)]_{VF})) - \xi([U, V]_{VF}, \pi^\#(\eta)]_{VF})) - (-(-\xi([V, X_{\eta(U)}]_{VF})) - \xi([V, \pi^\#(L_U(\eta))]_{VF})) \\
& U(V(\eta(\pi^\#(\xi)))) - V(U(\eta(\pi^\#(\xi)))) - [U, V]_{VF}(\eta(\pi^\#(\xi))) - \eta([U, [V, \pi^\#(\xi)]_{VF}]_{VF}) + \eta([V, [U, \pi^\#(\xi)]_{VF}]_{VF}) + \eta([U, V]_{VF}, \pi^\#(\xi)]_{VF}) + \xi([U, [V, \pi^\#(\eta)]_{VF}]_{VF}) \\
& U(V(\eta(\pi^\#(\xi)))) - V(U(\eta(\pi^\#(\xi)))) - [U, V]_{VF}(\eta(\pi^\#(\xi))) + \xi([U, [V, \pi^\#(\eta)]_{VF}]_{VF}) - \xi([V, [U, \pi^\#(\eta)]_{VF}]_{VF}) - \xi([U, V]_{VF}, \pi^\#(\eta)]_{VF}) \rightarrow U(V(\eta(\pi^\#(\xi)))) \\
& U(V(\eta(\pi^\#(\xi)))) - V(U(\eta(\pi^\#(\xi)))) - [U, V]_{VF}(\eta(\pi^\#(\xi))) \rightarrow [U, V]_{VF}(\eta(\pi^\#(\xi))) - [U, V]_{VF}(\eta(\pi^\#(\xi))) \quad [X(Y(f)) - Y(X(f)) = [X, Y]_{VF}(f) \text{ [permisive]]}] \text{ (axiom)} \\
& [U, V]_{VF}(\eta(\pi^\#(\xi))) - [U, V]_{VF}(\eta(\pi^\#(\xi))) \rightarrow 0 \quad [\text{simplify}] - \text{canonical-form pipeline (intra-loop)}
\end{aligned}$$

#### 1.4.2 (2') $\tilde{\mathcal{L}}_{d\iota_U \eta} V = -[\tilde{d}\tilde{\iota}_\eta U, V]_{\text{SN}}$

(2)'nin doğrudan duali:  $\tilde{\mathcal{L}}$  rolünde Lichnerowicz açılımı, SN bracket  $V$  ile alınıyor. Lhs ve rhs her iki tarafta da Hamilton vektör alanı  $X_{\eta(U)}$ 'ya iniyor — biri (lhs) SnBracket-on-1-forms üzerinden HVF (sign  $-$ ), diğeri (rhs)  $\tilde{d}_\pi$  uygulamasından pozitif  $\pi^\# \circ d - \text{Act}(\tilde{d}_\pi, f) \rightarrow \text{Neg}(\text{HVF}(f))$  köprüsü ikisini birleştiriyor.

```

[7]: head_lhs = Act(default_d, Act(interior(U), eta))
lhs = Act(tilde_lie(head_lhs, pi), V)
head_rhs = Act(tilde_d(pi), Act(tilde_interior(eta), U))
rhs = Neg(BracketApply(default_sn, head_rhs, V))
chain = prob.prove_derivator(lhs, rhs, eval_args=(xi,), side="multivector")
print(f"(2') multivec, Ltilde_{{d iota_U eta}} V = -[dtilde iotatilde_eta U, U
  ↪ V]_SN --> {len(chain)} adimda kapandı")
display_chain(chain)

```

(2') multivec, Ltilde\_{d iota\_U eta} V = -[dtilde iotatilde\_eta U, V]\_SN -->  
 25 adimda kapandi

[7]:

$$\begin{aligned}
 & \tilde{\mathcal{L}}_{d(\iota_U(\eta))}(V) \rightarrow -X_{d(\iota_U(\eta))}(V) + \pi\sharp(L_V(d(\iota_U(\eta)))) - [V, \pi\sharp(d(\iota_U(\eta)))]_{VF} \quad [\tilde{L}_-\alpha(U) = -X_-\alpha(U) + \pi\sharp(L_-U \alpha) - [U, \pi\sharp \alpha]_{VF} [\pi]] \text{ (axiom)} \\
 & L_V(d(\iota_U(\eta))) \rightarrow d(L_V(\iota_U(\eta))) \quad [L_-X \circ d = d \circ L_-X \text{ (tilde, any mode)}] \text{ (axiom)} \\
 & \pi\sharp(d(L_V(\iota_U(\eta)))) \rightarrow X_{L_V(\iota_U(\eta))} \quad [\pi\sharp(df) = X_-f [\pi]] \text{ (axiom)} \\
 & [V, \pi\sharp(d(\iota_U(\eta)))]_{VF} \rightarrow [V, X_{\eta(U)}]_{VF} \quad [\text{bare-atom-slot lift}] \text{ (axiom)} \\
 & (-X_{d(\iota_U(\eta))}(V) + X_{L_V(\iota_U(\eta))} - [V, X_{\eta(U)}]_{VF})(\xi) \rightarrow -X_{d(\iota_U(\eta))}(V)(\xi) + X_{L_V(\iota_U(\eta))}(\xi) - [V, X_{\eta(U)}]_{VF}(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
 & X_{d(\iota_U(\eta))}(V)(\xi) \rightarrow \xi(X_{d(\iota_U(\eta))}(V)) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
 & \xi(X_{d(\iota_U(\eta))}(V)) \rightarrow -(\pi\sharp(\xi))((d(\iota_U(\eta)))(V)) \quad [\alpha(X_-f) = -\pi\sharp(\alpha)(f) [\pi]] \text{ (axiom)} \\
 & (d(\iota_U(\eta)))(V) \rightarrow V(\iota_U(\eta)) \quad [d \text{ intrinsic (Koszul): } (d\omega)(X_-0, \dots, X_-p) = \sum \pm X_-i(\omega(\dots, \text{hat}_i, \dots)) + \sum \pm \omega([X_-i, X_-j]_{VF}, \dots)] \text{ (axiom)} \\
 & V(\iota_U(\eta)) \rightarrow V(\eta(U)) \quad [\text{bare } \iota_-X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ [permissive]} \text{ (axiom)} \\
 & X_{L_V(\iota_U(\eta))}(\xi) \rightarrow \xi(X_{L_V(\iota_U(\eta))}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
 & \xi(X_{L_V(\iota_U(\eta))}) \rightarrow -(\pi\sharp(\xi))(L_V(\iota_U(\eta))) \quad [\alpha(X_-f) = -\pi\sharp(\alpha)(f) [\pi]] \text{ (axiom)} \\
 & (\pi\sharp(\xi))(L_V(\iota_U(\eta))) \rightarrow (\pi\sharp(\xi))(V(\iota_U(\eta))) \quad [\text{Act}(D, L_-X(f)) \rightarrow \text{Act}(D, X(f)) \text{ [D vf-like, f 0-form]}] \text{ (axiom)} \\
 & V(\iota_U(\eta)) \rightarrow V(\eta(U)) \quad [\text{bare } \iota_-X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X))] \text{ [permissive]} \text{ (axiom)} \\
 & [V, X_{\eta(U)}]_{VF}(\xi) \rightarrow \xi([V, X_{\eta(U)}]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
 & \tilde{d}(\tilde{\iota}_\eta(U)) \rightarrow \tilde{d}(U(\eta)) \quad [\text{bare } \tilde{\iota}_-\omega(V) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(V, \omega)) \text{ (V deg 1)}] \text{ (axiom)} \\
 & U(\eta) \rightarrow \eta(U) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
 & \tilde{d}(\eta(U)) \rightarrow -X_\eta(U) \quad [\text{Act}(\tilde{d}_-\pi, f) \rightarrow -X_-f \text{ [geometer sign]}] \text{ (axiom)} \\
 & [-X_\eta(U), V]_{[\cdot, \cdot]_{SN}} \rightarrow -[X_\eta(U), V]_{[\cdot, \cdot]_{SN}} \quad [(-a), b]_{SN} \rightarrow -[a, b]_{SN} / [a, (-b)]_{SN} \rightarrow -[a, b]_{SN}] \text{ (axiom)} \\
 & [X_\eta(U), V]_{[\cdot, \cdot]_{SN}} \rightarrow [X_\eta(U), V]_{VF} \quad [[U, V]_{SN} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\
 & [X_\eta(U), V]_{VF} \rightarrow -[V, X_\eta(U)]_{VF} \quad [[Y, X]_{VF} \rightarrow -[X, Y]_{VF} \text{ [repr-canonical]}] \text{ (axiom)} \\
 & (-(-(-[V, X_\eta(U)]_{VF}))) (\xi) \rightarrow -(-(-[V, X_\eta(U)]_{VF})) (\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
 & (-(-[V, X_\eta(U)]_{VF})) (\xi) \rightarrow -(-[V, X_\eta(U)]_{VF}) (\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
 & (-[V, X_\eta(U)]_{VF}) (\xi) \rightarrow -[V, X_\eta(U)]_{VF} (\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\
 & [V, X_\eta(U)]_{VF} (\xi) \rightarrow \xi([V, X_\eta(U)]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\
 & (-(-(\pi\sharp(\xi))(V(\eta(U)))) - (\pi\sharp(\xi))(V(\eta(U))) - \xi([V, X_\eta(U)]_{VF})) - (-(-(-\xi([V, X_\eta(U)]_{VF})))) \rightarrow 0 \quad [\text{simplify}] \text{ --- canonical-form pipeline (intra-}
 \end{aligned}$$

**1.4.3 (3')**  $0 = \tilde{d}\tilde{\iota}_{\mathcal{L}U\eta} V - \tilde{d}\tilde{\iota}_{d\iota_V\eta} U + \tilde{d}\tilde{\iota}_\eta [U, V]_{SN}$

(3)'ün doğrudan duali: aynı şekilde sıfır-toplam, ilk iki terim lhs<sub>pair</sub>, üçüncü (pozitif) terim rhs<sub>third</sub> olarak.

[8]:

```

v1p = Act(lie_derivative(U), eta)
v2p = Act(default_d, Act(interior(V), eta))
bracket_uv = BracketApply(default_sn, U, V)

lhs_pair = Sum(
    Act(tilde_d(pi), Act(tilde_interior(v1p), V)),
    Act(tilde_d(pi), Act(tilde_interior(eta), bracket_uv)),
)
rhs_third = Act(tilde_d(pi), Act(tilde_interior(v2p), U))
chain = prob.prove_derivator(lhs_pair, rhs_third, eval_args=(xi,),
    ↪side="multivector")

```

```
print(f"(3') multivec, zero-sum reformulated lhs_pair = rhs_third --> 23  
      ↳{len(chain)} adimda kapandi")  
display_chain(chain)
```

(3') multivec, zero-sum reformulated lhs\_pair = rhs\_third --> 23 adimda  
kapandi

[8]:

$$\begin{aligned} \tilde{d}(\tilde{\iota}_{L_U(\eta)}(V)) &\rightarrow \tilde{d}(V(L_U(\eta))) \quad [\text{bare } \tilde{\iota}_\omega(V) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(V, \omega)) \text{ (V deg 1)}] \text{ (axiom)} \\ V(L_U(\eta)) &\rightarrow (L_U(\eta))(V) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ (L_U(\eta))(V) &\rightarrow U(\eta(V)) - \eta([U, V]_{VF}) \quad [L\_X \text{ intrinsic: } (L\_X \omega)(Y\_1, \dots) = X(\omega(Y\_1, \dots)) - \Sigma \omega(\dots, [X, Y\_i]_{VF}, \dots)] \text{ (axiom)} \\ \tilde{d}(U(\eta(V)) - \eta([U, V]_{VF})) &\rightarrow -X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))} \quad [\text{Act}(\tilde{d}_\pi, f) \rightarrow -X\_f \text{ [geometer sign]}] \text{ (axiom)} \\ [U, V]_{[\cdot, \cdot]_{SN}} &\rightarrow [U, V]_{VF} \quad [[U, V]_{SN} \rightarrow [U, V]_{VF} \text{ [both 1-vectors]}] \text{ (axiom)} \\ \tilde{d}(\tilde{\iota}_\eta([U, V]_{VF})) &\rightarrow \tilde{d}([U, V]_{VF}(\eta)) \quad [\text{Act}(D, \tilde{\iota}_\alpha([U, V]_{VF})) \rightarrow \text{Act}(D, \text{MultiEval}([U, V]_{VF}, \alpha, \text{covector}))] \text{ (axiom)} \\ [U, V]_{VF}(\eta) &\rightarrow \eta([U, V]_{VF}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ \tilde{d}(\eta([U, V]_{VF})) &\rightarrow -X_{\eta([U, V]_{VF})} \quad [\text{Act}(\tilde{d}_\pi, f) \rightarrow -X\_f \text{ [geometer sign]}] \text{ (axiom)} \\ (-X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))} - X_{\eta([U, V]_{VF})})(\xi) &\rightarrow -X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))}(\xi) - X_{\eta([U, V]_{VF})}(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\ X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))}(\xi) &\rightarrow \xi(X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ \xi(X_{(U(\eta(V)) + (-\eta([U, V]_{VF})))}) &\rightarrow -(\pi^\sharp(\xi))(U(\eta(V)) - \eta([U, V]_{VF})) \quad [\alpha(X\_f) = -\pi^\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\ X_{\eta([U, V]_{VF})}(\xi) &\rightarrow \xi(X_{\eta([U, V]_{VF})}) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ \xi(X_{\eta([U, V]_{VF})}) &\rightarrow -(\pi^\sharp(\xi))(\eta([U, V]_{VF})) \quad [\alpha(X\_f) = -\pi^\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\ \tilde{d}(\tilde{\iota}_{d_{\iota_V}(\eta)}(U)) &\rightarrow \tilde{d}(U(d_{\iota_V}(\eta))) \quad [\text{bare } \tilde{\iota}_\omega(V) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(V, \omega)) \text{ (V deg 1)}] \text{ (axiom)} \\ U(d_{\iota_V}(\eta)) &\rightarrow (d_{\iota_V}(\eta))(U) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ (d_{\iota_V}(\eta))(U) &\rightarrow U(\iota_V(\eta)) \quad [d \text{ intrinsic (Koszul): } (d\omega)(X\_0, \dots, X\_p) = \Sigma \pm X\_i(\omega(\dots, \hat{\omega}_i, \dots)) + \Sigma \pm \omega([X\_i, X\_j]_{VF}, \dots)] \text{ (axiom)} \\ U(\iota_V(\eta)) &\rightarrow U(\eta(V)) \quad [\text{bare } \iota\_X(\omega) \text{ inside } \text{Act}(D, \_) \rightarrow \text{Act}(D, \text{MultiEval}(\omega, X)) \text{ [permissive]}] \text{ (axiom)} \\ \tilde{d}(U(\eta(V))) &\rightarrow -X_U(\eta(V)) \quad [\text{Act}(\tilde{d}_\pi, f) \rightarrow -X\_f \text{ [geometer sign]}] \text{ (axiom)} \\ (-X_U(\eta(V)))(\xi) &\rightarrow -X_U(\eta(V))(\xi) \quad [\text{MultiEval head linearity}] \text{ (axiom)} \\ X_U(\eta(V))(\xi) &\rightarrow \xi(X_U(\eta(V))) \quad [X(\omega) \rightarrow \omega(X) \text{ [arity-1 covector pairing flip]}] \text{ (axiom)} \\ \xi(X_U(\eta(V))) &\rightarrow -(\pi^\sharp(\xi))(U(\eta(V))) \quad [\alpha(X\_f) = -\pi^\sharp(\alpha)(f) \text{ } [\pi]] \text{ (axiom)} \\ (-(-(\pi^\sharp(\xi))(U(\eta(V)) - \eta([U, V]_{VF}))) - (-(\pi^\sharp(\xi))(\eta([U, V]_{VF})))) &\rightarrow (-(-(\pi^\sharp(\xi))(U(\eta(V))) - (\pi^\sharp(\xi))(\eta([U, V]_{VF})))) \rightarrow 0 \quad [\text{simplify}] \text{ — canonical-form pipeli} \end{aligned}$$

## 1.5 Sonuç

Altı §3.1.5 özdeşliği — form (1)/(2)/(3) ve dual (1')/(2')/(3') — tek `prob.prove_derivator(...)` çağrısıyla syntactic olarak kapandı. Adım sayıları: 109 / 21 / 30 (form) ve 117 / 25 / 23 (multivec-tor).

**Form taraf:**

- (1):  $D_{\mathcal{L}_U}^{T^*M}(\eta, \mu) = \mathcal{L}_{\tilde{K}_\eta U} \mu + K_{\tilde{K}_\mu U} \eta$  — eval ( $Y$ )
- (2):  $\mathcal{L}_{\tilde{d}_U \eta} \mu = -[d\iota_U \eta, \mu]_K$  — eval ( $Y$ )
- (3):  $\sum d\iota_\bullet = 0$  — eval ( $Y$ )

**Multivektör taraf:**

- (1'):  $\tilde{D}_{\tilde{\mathcal{L}}_\eta}^{\text{SN}}(U, V) = \tilde{\mathcal{L}}_{K_U \eta} V + \tilde{K}_{K_V \eta} U$  — eval ( $\xi$ )
- (2'):  $\tilde{\mathcal{L}}_{d_U \eta} V = -[\tilde{d}_U \eta, V]_{\text{SN}}$  — eval ( $\xi$ )
- (3'):  $\sum \tilde{d}_\bullet = 0$  — eval ( $\xi$ )

Faz 15.C şu üç adımı sahneye getirmek zorunda kaldı: form-side Cartan-magic + tilde Lichnerowicz açılımı (vektör alanı için),  $\pi^\sharp$  Neg-doğrusallığı, ve `LieBracketVF` gibi opak operator atomlarının slot-lift'i. Bu üçü olmaksızın özellikle (1') 117 adımlık kapanışına ulaşamıyor — `display_chain(chain)` ile her adım `Definition` etiketiyle birlikte LaTeX olarak okunabilir.